Algebra Review: Unit 1 — Expressions & Equations

Answer Key

Station 1: Evaluating and Simplifying Expressions

1. 21 $\quad 3(4)+4+5$
2. 5 $\quad \dfrac{18+2}{4}$
3. 23 $\quad 12+16-5$
4. $13x-12$
5. $4x+14$
6. $-7y+3$
7. $7x-3$
8. $3a+9b$
9. 13 $\quad \sqrt{25+144}=\sqrt{169}$
10. $3n-5$

Station 2: One- and Two-Step Equations

1. $x=-5$
2. $y=4$
3. $n=-7$
4. $m=-15$
5. $x=5$
6. $y=6$
7. $m=18$
8. $n=9$
9. $x=11$
10. $12+5v=47;\ v=7$ visits

Station 3: Multi-Step Equations and Variables on Both Sides

1. $x=5$
2. $x=5$
3. $y=6$
4. $x=4$
5. $x=5$
6. $x=12$
7. $x=14$
8. no solution
9. all real numbers
10. $x=-5$

Station 4: Literal Equations and Word Problems

1. $r=\dfrac{C}{2\pi}$
2. $h=\dfrac{3V}{\pi r^2}$
3. $x=\dfrac{y-b}{m}$
4. $w=\dfrac{P-2\ell}{2}$
5. $b=\dfrac{2A}{h}-a$
6. $y=\dfrac{12-3x}{4}$
7. $3n-8=n+12;\ n=10$
8. 28, 29, 30 $\quad n+(n+1)+(n+2)=87$
9. width $=5$ cm, length $=13$ cm $\quad 2(2w+3)+2w=36$
10. $20+0.10m=5+0.25m;\ m=100$ minutes